The Promise and Practice of EdTech in Africa: A Review of Learning Outcomes

Research Internship Task 1 : Priscilla Oforiwaa Opoku-Yeboah

October 3 , 2026

Introduction

Education systems across Africa face persistent challenges, including limited access to quality learning materials, teacher shortages, and infrastructure gaps. Technology-Enhanced Learning (TEL) has emerged as a potential solution to these challenges, promising to bridge educational divides and improve learning outcomes. However, the integration of educational technology (EdTech) in African contexts is complex. This review examines recent literature on EdTech in Africa, focusing on its actual impact on learning outcomes, the challenges of implementation, and the critical gap between access and effective pedagogical use.

Evidence of Impact

A growing body of evidence suggests that when implemented correctly, EdTech can significantly improve learning outcomes. A notable randomized controlled trial in Ghana evaluated the effectiveness of interactive satellite-transmitted instruction in rural primary schools (Johnston & Ksoll, 2022). The study found substantial learning gains in both mathematics and English, demonstrating that distance learning technologies can overcome geographical barriers. Similarly, mobile learning and AI-driven tutoring systems have shown promise. For instance, recent studies suggest that AI tutoring tools, accessed via basic mobile phones and messaging apps like WhatsApp, can produce learning gains equivalent to an extra year of schooling at a very low marginal cost (Africa Practice, 2026).

The Usage Gap: Substitution vs. Redefinition

While the potential is high, a major theme in the literature is the gap between the availability of technology and its effective use in the classroom. A systematic review of 199 articles covering 27 years of TEL in African higher education found that while technology has increased access to education, its pedagogical application remains limited (Habarurema et al., 2024). Using the SAMR framework, the authors found that most technology use in Africa is at the "substitution" (doing the same task with new tools)

and “augmentation” levels. Deep integration at the “modification” and “redefinition” levels—where technology fundamentally transforms the learning process—is rare. This highlights a critical need for continuous professional development for educators to move beyond basic usage.

Challenges and Limitations

The literature also highlights significant barriers to the successful deployment of EdTech in Africa. Infrastructure remains a primary concern, including unreliable electricity, lack of internet connectivity, and a shortage of devices (World Bank, 2023). Furthermore, not all technological interventions are successful. A randomized control trial in Sierra Leone found that live tutoring calls conducted during the COVID-19 pandemic did not improve learning outcomes, underscoring that EdTech must be carefully designed to meet specific contextual needs (Beuermann et al., 2023). The digital divide remains a persistent threat, with marginalized communities at risk of being left further behind if EdTech policies do not prioritize equity and local language inclusion (Citi Newsroom, 2026).

Conclusion

The literature confirms that EdTech holds transformative potential for African education, particularly through mobile learning and AI. However, to realize the “two gains” of access and pedagogical improvement, interventions must move beyond basic substitution. Future efforts must focus on teacher training, robust policy frameworks, and addressing systemic infrastructure challenges to ensure that technology serves as an equitable tool for learning rather than a source of further division.

References

- Beuermann, D. W., et al. (2023). *Live tutoring calls did not improve learning during the COVID-19 pandemic in Sierra Leone.* Journal of Development Economics.
- Habarurema, J. B., et al. (2024). Technology-Enhanced Learning in Africa: A Challenge and Opportunity? *Africa Education Review*, 1-20. `https://doi.org/10.1080/18146627.2024.2433982`
- Johnston, J., & Ksoll, C. (2022). *Effectiveness of interactive satellite-transmitted instruction: Experimental evidence from Ghanaian primary schools.* Journal of Development Economics.
- World Bank. (2023). *The World Bank and EdTech in Sub-Saharan Africa: Evolving Priorities, Persistent Gaps.*
- Africa Practice. (2026). *AI Doesn't Need Perfect Classrooms.*
- Citi Newsroom. (2026). *From screens to strategy: Is Ghana finally ready to close the digital divide for every child?*